

My Files in the Sky

Session 1: An Introduction to Cloud Computing & Infrastructure

Target Grade: Grade 6 | **Subject:** Future Tech & Digital Literacy | **Provider:** Planrs Academy

Scenario: The Broken Phone Nightmare!

Imagine this: You have hundreds of precious memories saved on your phone — photos from birthday parties, family vacations, and funny videos with friends. One day... *SPLASH!* Your phone accidentally drops into a deep puddle of water. Your heart sinks. Is all your digital data lost forever?

The Solution: If your files were stored only on the physical device, they might be gone. But if they were saved in **The Cloud**, all your photos and documents are 100% safe! You can simply log in from a new phone or computer and access everything instantly.

1. What is the Cloud? (It's Not Weather!)

When computer scientists talk about "The Cloud," they do **NOT** mean a fluffy rain cloud in the sky. The Cloud is a global network of thousands of ultra-powerful computers (called **Servers**) located in huge secure buildings called **Data Centers**.

The Digital Bank Locker Analogy

Think of the Cloud as a **Digital Bank Locker**. Instead of keeping all your valuable gold or jewelry in a drawer at home (where it could get misplaced or damaged), you keep it protected in a high-security bank vault. You can walk into any bank branch in the world, enter your password/key, and access your locker. Similarly, the Cloud keeps your digital files safe in secure servers so you can open them from any computer or phone globally.

2. Where Does the Cloud Live in India?

Data centers are like *digital apartment complexes* where apps and databases live. In India, tech giants have established giant data centers equipped with round-the-clock cooling, backup power generators, and biometric security.

Amazon Web Services (AWS)

- **Role:** Pioneer of cloud computing.
- **Major Indian Data Centers:** Mumbai & Hyderabad.
- **Popular Apps Using AWS:** Netflix, SonyLIV, and top Indian startups.

◆ Microsoft Azure

- **Role:** Enterprise & education giant.
- **Major Indian Data Centers:** Pune, Chennai, & Mumbai.
- **Popular Apps Using Azure:** School management portals, MS Teams, and corporate systems.

3. Key Benefits of Cloud Computing

Superpower	How It Works	Real-Life Analogy
Accessibility	Access your files from any connected device (phone, tablet, laptop) anytime, anywhere.	Carrying your library card gives you access to books at any library branch.
Scalability	Storage and computing power can instantly grow or shrink based on user demand.	An Indian wedding caterer who can effortlessly serve 50 guests or scale up to 500 guests!
Data Safety	Automatic backups prevent data loss even if your physical device gets broken or stolen.	Keeping duplicate copies of important documents in a fireproof safety vault.

4. Real-World Cloud Apps You Use Everyday

You interact with cloud computing daily without even realizing it:

- **Disney+ Hotstar:** Start watching a movie on your TV at home. Pause it, board a car, and resume watching from the exact same minute on your mobile phone! The Cloud tracks your viewing progress across devices.
- **Paytm & Google Pay:** Your money balance and transaction histories live securely on cloud servers, not inside your physical phone hardware.
- **BYJU'S & Google Classroom:** Homework assignments, lecture videos, and quizzes are hosted in cloud storage for millions of students.
- **Spotify & Wynk Music:** Millions of songs stream directly from remote cloud servers down to your headphones in real time.

Student Worksheet & Skill Activity

Session 1 • Hands-on Cloud vs. Local Storage Exploration

Activity 1: Pack Your Digital Suitcase!

Imagine you are going on a trip and packing a digital suitcase. Some items must stay tied to your local device hardware, while others should travel with you in the Cloud so you can access them anywhere.

Item Bank:

Family Vacation Photos

Computer Graphic Card Drivers

Gmail Emails

Device Operating System (Windows/iOS)

Saved Mobile Games (Cloud Sync)

Keyboard/Mouse Hardware Files

Google Docs Homework

Monitor Display Settings

Directions: Classify each item from the bank into its correct category below.

Cloud Suitcase (Travels Everywhere)

Local Device (Stays On Hardware)

Activity 2: Cloud Real-World Scenario Analysis

Read the real-world situations below and fill in your answers:

Scenario A: Rohan is working on a Science presentation in Google Slides on his school desktop. The school bell rings before he finishes. Can Rohan complete his work at home on his mother's laptop? Explain how the cloud makes this possible.

Scenario B: An online shopping app receives 1,000 visitors per day during normal months, but during Diwali sale week, 100,000 visitors buy items at once. Which cloud superpower allows the website to handle this sudden surge without crashing?

Activity 3: True or False Challenge

1. The Cloud requires actual water vapor and rain clouds to operate. []
2. AWS and Microsoft Azure own giant server buildings called Data Centers in India. []
3. If your tablet breaks into pieces, all files stored in Google Drive are destroyed forever. []
4. Scalability means the cloud can automatically adjust storage size as your data grows. []

Practice Questions & Knowledge Check

Review Assessment for Grade 6 Students • Cloud Computing Fundamentals

Instructions: Answer the following practice questions based on the lesson module. Write your answers neatly on a separate sheet of paper or in your computer science notebook.

Section A: Multiple Choice Questions (1 Mark Each)

Q1. What is the main hardware infrastructure that powers cloud computing?

1. Satellites floating in the atmosphere
2. Large networks of server computers located in Data Centers
3. Cell phone towers situated on rooftops
4. Wi-Fi routers placed inside households

Q2. Which cloud benefit allows an app like Hotstar to seamlessly handle millions of viewers during an IPL match?

1. Accessibility
2. Waterproofing
3. Scalability
4. Local Storage

Q3. In which of the following Indian cities can you find major data centers operated by AWS or Microsoft Azure?

1. Mumbai & Hyderabad
2. Shimla & Darjeeling
3. Srinagar & Leh
4. Ooty & Gangtok

Q4. Which of these items is typically stored on a local physical device rather than in the cloud?

1. Google Drive spreadsheet
2. Hardware graphics card driver
3. Netflix movie library
4. Email messages in Gmail

Section B: Short Answer Questions (2–3 Marks Each)

Q5. Explain the "**Digital Bank Locker**" analogy in your own words. Why is cloud storage safer than saving files on a single USB flash drive?

Q6. Name the two major cloud service providers (often called "digital landlords") discussed in the lesson, along with the parent companies behind them.

Q7. How does cloud synchronization allow you to pause a movie or song on one device and continue playing it from the exact same point on another device?

Section C: Critical Thinking & Application (5 Marks Each)

Q8. Imagine you are an entrepreneur launching a new educational gaming app in India.

- (a) Why would renting cloud server space from AWS or Azure be better than buying physical computer servers for your home office?
- (b) How does the concept of "Scalability" help your business save money when starting out?

Q9. Think about a school library vs. a personal bookshelf. Draw a comparison showing how a school library is similar to Cloud Computing, while a personal bookshelf is like Local Device Storage. Mention at least two points of comparison.